

A Nondense Diagonal Counterexample to the Fractional-Power Limit in Remark 2.11

Candidate counterexample packet for arXiv:2006.07008

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Status. Candidate full counterexample to the open problem as stated; likely valid, pending expert review. The example uses a non-densely defined positive operator, which the source paper explicitly permits. It does not settle a separately imposed dense-domain variant.

Abstract

Batty and Chen ask whether their Komatsu–Chen domain-characterization limit holds for the missing exponents $\alpha = n + it$. On ℓ^∞ , let A be the diagonal operator with eigenvalues k . For $x_k = k^{-\alpha}$ and $\beta = \alpha + 1$, one has $A^\alpha x = \mathbf{1}$, but the normalized truncation at N is exactly $((N/(N+k))^\alpha)_k$. These truncations lie in the closed subspace c_0 and converge coordinatewise to $\mathbf{1} \notin c_0$, so they do not converge in ℓ^∞ . In fact, the same counterexample works for every $\operatorname{Re} \alpha > 0$.

1 Source question and scope

The source is C. Batty and C. Chen, *Besov spaces associated with non-negative operators on Banach spaces*, arXiv:2006.07008v1 [1]. Formula (2.37) on source PDF p. 15 is the limit under discussion:

2.10] characterized the fractional domain $D(A^\alpha)$ for the case in which $\overline{D(A)} = X$ and $\alpha \in \mathbb{C}_+$ with $\operatorname{Re} \alpha \notin \mathbb{N}$.

Let A be non-negative on X and let $\alpha \in \mathbb{C}_+$ with $\operatorname{Re} \alpha < n \in \mathbb{N}$. Recall that (2.11) admits a more general version. More precisely, on the one hand, the regularity of x in (2.11) can be weakened from $x \in D(A^n)$ to $x \in X$ such that there exists the limit $\lim_{N \rightarrow \infty} \int_0^N \lambda^\alpha A^n (\lambda + A)^{-n} x \frac{d\lambda}{\lambda}$ (see [51] Theorem 2.10]). On the other hand, (2.11) can be extended from integers $n > \operatorname{Re} \alpha$ to complex numbers β satisfying $\operatorname{Re} \beta > \operatorname{Re} \alpha$, as shown in [65] Corollary 5.1.13, (5.17)]. For the sake of the reader's convenience, we reformulate [65] Corollary 5.1.13] as follows.

Lemma 2.7. *Let A be non-negative on X and let $\alpha, \beta \in \mathbb{C}_+$ with $\operatorname{Re} \alpha < \operatorname{Re} \beta$. Fix $x \in X$ such that there exists the limit*

$$(2.37) \quad \lim_{N \rightarrow \infty} \frac{\Gamma(\beta)}{\Gamma(\alpha)\Gamma(\beta - \alpha)} \int_0^N \lambda^\alpha A^\beta (\lambda + A)^{-\beta} x \frac{d\lambda}{\lambda} := y.$$

Then $x \in D(A^\alpha)$ and $A^\alpha x = y$.

In particular, the limit (2.37) exists for $x \in X$ satisfying

$$(2.38) \quad \int_0^\infty \|\lambda^\alpha A^\beta (\lambda + A)^{-\beta} x\| \frac{d\lambda}{\lambda} < \infty.$$

Thus, we have the following consequence of Lemma 2.7

Corollary 2.8. *Let A be non-negative on X and let $\alpha, \beta \in \mathbb{C}_+$ with $\operatorname{Re} \alpha < \operatorname{Re} \beta$. Fix $x \in X$ such that (2.38) holds. Then $x \in D(A^\alpha)$ and*

$$(2.39) \quad A^\alpha x = \frac{\Gamma(\beta)}{\Gamma(\alpha)\Gamma(\beta - \alpha)} \int_0^\infty \lambda^\alpha A^\beta (\lambda + A)^{-\beta} x \frac{d\lambda}{\lambda}.$$

By use of (2.1) and (2.3), it is easy to verify that (2.38) is satisfied whenever $x \in D(A^{\alpha+\epsilon})$. Thus, we have the following simple consequence of Corollary 2.8. Also see [25] Lemma 3.1].

Corollary 2.9. *Let A be non-negative on X , and let $\alpha, \beta \in \mathbb{C}_+$ with $\operatorname{Re} \alpha < \operatorname{Re} \beta$. Then (2.39) holds for $x \in D(A^{\alpha+\epsilon})$ with $\epsilon > 0$.*

On source PDF p. 16, Lemma 2.10 asserts the converse representation except possibly when $\operatorname{Re} \alpha \in \mathbb{N}$ and $\operatorname{Im} \alpha \neq 0$, and Remark 2.11 asks whether it remains true for $\alpha = n + it$:

Compared with Lemma 2.7 above, the following lemma reads that the limit (2.37) exists whenever $x \in D(A^\alpha)$ with $\alpha \in \mathbb{C}_+$, possibly except for the case in which $\operatorname{Re} \alpha \in \mathbb{N}$ and $\operatorname{Im} \alpha \neq 0$, as shown in [25] Theorem 3.5, (i) $\Rightarrow$ (iii)].

Lemma 2.10. *Let A be non-negative on X and let $\alpha, \beta \in \mathbb{C}_+$ with $\operatorname{Re} \alpha < \operatorname{Re} \beta$. If $x \in D(A^\alpha)$, then there exists the limit (2.37) and $A^\alpha x = y$, possibly except for the case in which $\operatorname{Re} \alpha \in \mathbb{N}$ and $\operatorname{Im} \alpha \neq 0$.*

Remark 2.11. Here we say a few more words on Lemma 2.10. Recall that, as for the three statements (i), (ii) and (iii) given in [25] Theorem 3.5], the equivalence (i) $\Leftrightarrow$ (ii) and the implication (iii) $\Rightarrow$ (i) both hold, while the implication (i) $\Rightarrow$ (iii) was merely verified for the case in which $\operatorname{Re} \alpha \notin \mathbb{N}$ or $\alpha = n \in \mathbb{N}$ (since, in Step II of the proof given in [25] Theorem 3.5], the use of integration by parts is possible merely in the case $\operatorname{Re} \alpha = 1$ with $\operatorname{Im} \alpha = 0$, i.e., $\alpha = 1$). Thus, whether Lemma 2.10 above holds in the case $\alpha = n + it$ with $n \in \mathbb{N}$ and $0 \neq t \in \mathbb{R}$ is still an **open problem**. We also refer the reader to [51] Theorem 2.10] and [65] Theorem 6.1.3 and Remark 6.1.1] for more information on this topic.

Thus, with $\alpha, \beta \in \mathbb{C}_+$ and $\operatorname{Re} \alpha < \operatorname{Re} \beta$, the question is whether every $x \in D(A^\alpha)$ satisfies

$$A^\alpha x = \lim_{N \rightarrow \infty} \frac{\Gamma(\beta)}{\Gamma(\alpha)\Gamma(\beta - \alpha)} \int_0^N \lambda^\alpha A^\beta (\lambda + A)^{-\beta} x \frac{d\lambda}{\lambda} \quad (1)$$

in the case $\alpha = n + it$, $n \in \mathbb{N}$, $t \in \mathbb{R} \setminus \{0\}$.

The nondense-domain scope is essential here. The paper says explicitly in its introduction that its general non-negative operators need not be densely defined or have dense range (source PDF p. 2):

1. INTRODUCTION

This paper is devoted to the theory of Besov spaces from the operator-theoretic point of view and extending some recent work on Besov spaces associated with operators (see, for example, Besov spaces associated with heat kernels [17] or sectorial operators [37] or, in particular, sectorial operators of angle zero [57]). A new point in our development of the Besov space $B_{q,X}^{s,A}$ associated with an operator A is that we allow the operator A to be a general non-negative operator (not necessarily defined densely or with dense range) on a Banach space X and allow the space $B_{q,X}^{s,A}$ to admit a full range of the smoothness index $s \in \mathbb{R}$ and size index $0 < q \leq \infty$.

The story originates from the theory of fractional powers of operators. Roughly speaking, the concept of fractional powers of operators is a continuous analogue of the discrete scale of regularity associated with operators in the theory of abstract

2 Proof intuition

For a nondensely defined operator, fractional powers can map a vector in their domain to a point outside $\overline{D(A)}$. In contrast, every truncated resolvent integral in (1) remains inside $\overline{D(A)}$. A diagonal operator on ℓ^∞ makes this mismatch exact: $\overline{D(A)} = c_0$, while a carefully chosen x has $A^\alpha x = \mathbf{1}$.

Taking $\beta = \alpha + 1$ collapses the incomplete beta integral to the elementary antiderivative $(u/(1+u))^\alpha$, so there are no approximation or functional-calculus ambiguities.

3 Counterexample

We use principal complex powers of positive real numbers.

Theorem 1. *Let $\alpha \in \mathbb{C}$ satisfy $\operatorname{Re} \alpha > 0$ and set $\beta = \alpha + 1$. There are a complex Banach space X , a positive operator A on X , and $x \in D(A^\alpha)$ for which the limit in (1) does not exist. Consequently, Remark 2.11 has a negative answer in the generality stated in the source paper. In particular this applies to every $\alpha = n + it$ with $n \in \mathbb{N}$ and $t \neq 0$.*

Proof. Take $X = \ell^\infty(\mathbb{N})$ and define

$$D(A) = \{z = (z_k)_{k \geq 1} \in \ell^\infty : (kz_k)_{k \geq 1} \in \ell^\infty\}, \quad (Az)_k = kz_k.$$

Operator hypotheses. The operator is closed: if $z^{(j)} \rightarrow z$ and $Az^{(j)} \rightarrow w$ in ℓ^∞ , then coordinate evaluation gives $w_k = kz_k$, whence $(kz_k) = w \in \ell^\infty$. For $\lambda > 0$,

$$((\lambda + A)^{-1}z)_k = \frac{z_k}{\lambda + k}, \quad \|\lambda(\lambda + A)^{-1}\| = \sup_{k \geq 1} \frac{\lambda}{\lambda + k} \leq 1.$$

This resolvent maps X onto $D(A)$. Moreover $0 \in \rho(A)$ and A^{-1} is multiplication by k^{-1} . Thus A is positive, hence non-negative, in the terminology of the source. Also

$$D(A) \subset c_0, \quad c_{00} \subset D(A), \quad \overline{D(A)} = c_0.$$

Fractional powers. Put $B = A^{-1}$. The bounded operator B is diagonal with entries k^{-1} . The Balakrishnan–Komatsu formula for a bounded non-negative operator, applied coordinatewise, gives

$$(B^\alpha z)_k = k^{-\alpha} z_k.$$

Indeed, for any integer $m > \operatorname{Re} \alpha$, its scalar integral is the usual beta integral

$$\frac{\Gamma(m)}{\Gamma(\alpha)\Gamma(m-\alpha)} \int_0^\infty \lambda^{\alpha-1} \frac{k^{-m}}{(\lambda + k^{-1})^m} d\lambda = k^{-\alpha},$$

and the operator-valued integral converges in norm. By the source's definition for a positive unbounded operator, $A^\alpha = (B^\alpha)^{-1}$. Hence

$$D(A^\alpha) = \{z \in \ell^\infty : (k^\alpha z_k)_k \in \ell^\infty\}, \quad (A^\alpha z)_k = k^\alpha z_k.$$

The witness and the exact truncation. Let $x_k = k^{-\alpha}$. Since $\operatorname{Re} \alpha > 0$, $x \in c_0 \subset X$; the preceding formula shows that $x \in D(A^\alpha)$ and $A^\alpha x = \mathbf{1}$. The same diagonal functional-calculus computation gives $((\lambda + A)^{-\beta} z)_k = (\lambda + k)^{-\beta} z_k$ and $(A^\beta z)_k = k^\beta z_k$ on its domain. Since $\beta = \alpha + 1$,

$$\frac{\Gamma(\beta)}{\Gamma(\alpha)\Gamma(\beta-\alpha)} = \alpha.$$

The k th coordinate of the normalized truncation $T_N x$ in (1) is therefore

$$\begin{aligned}(T_N x)_k &= \alpha \int_0^N \lambda^{\alpha-1} k(\lambda + k)^{-\alpha-1} d\lambda \\ &= \alpha \int_0^{N/k} u^{\alpha-1} (1 + u)^{-\alpha-1} du \\ &= \left(\frac{N}{N + k} \right)^\alpha,\end{aligned}$$

because

$$\frac{d}{du} \left(\frac{u}{1 + u} \right)^\alpha = \alpha u^{\alpha-1} (1 + u)^{-\alpha-1}.$$

The finite-cutoff Bochner integral is legitimate: after division by λ , its norm is $O(\lambda^{\operatorname{Re} \alpha - 1})$ near zero and the integrand is norm continuous away from zero.

Nonconvergence. For each finite N , $(T_N x)_k \rightarrow 0$ as $k \rightarrow \infty$, so $T_N x \in c_0$. For each fixed k , however, $(T_N x)_k \rightarrow 1$ as $N \rightarrow \infty$. If $(T_N x)$ converged in ℓ^∞ , the continuity of every coordinate evaluation would force its limit to be $\mathbf{1}$. But c_0 is norm closed and every $T_N x$ lies in c_0 , whereas $\mathbf{1} \notin c_0$. This contradiction proves that the limit does not exist. $\square$

4 Verification, scope, and limitations

The proof is entirely explicit. The main adversarial checks are:

1. $A : D(A) \rightarrow X$ is bijective although $D(A)$ is not dense; its inverse is bounded on all of X , exactly as required by $0 \in \rho(A)$.
2. The fractional-power computation follows from the same bounded-inverse definition used in the source, not from an unproved coordinatewise convention.
3. For every finite N , the integral is a genuine Bochner integral in ℓ^∞ ; only the improper limit fails.
4. The contradiction is norm-topological, not merely the observation that pointwise convergence need not be uniform.

The example actually disproves the analogue of Lemma 2.10 for every $\alpha \in \mathbb{C}_+$ if nondense domains are admitted. It does not address the variant under the additional hypothesis $\overline{D(A)} = X$. The 2016 Chen–Li paper cited by Batty–Chen is described as treating densely defined operators; no contradiction with that restricted theorem is claimed.

An optional script, `code/verifier.py`, compares numerical quadrature with the exact closed form for a sample nonreal exponent. This is only a sanity check; the antiderivative above is the proof.

5 Bounded literature and novelty check

Searches on 11 August 2026 covered the run’s registry, solution, attempt, and proof-gap indexes; the exact arXiv id and title; the phrases “mystery of imaginary powers,” “ $n + it$,” and “characterizations of domains of fractional powers”; the cited Chen–Li paper; title/citation variants; and combinations of non-densely defined fractional powers with diagonal operators on ℓ^∞ and c_0 . Background literature on fractional powers of non-densely defined operators was found, but no later paper explicitly answering Remark 2.11 and no matching diagonal counterexample was located. Novelty is therefore provisional.

6 Human-review recommendation

Verify first that Remark 2.11 is intended in the paper’s advertised general framework, rather than with an unstated dense-domain restriction. Then check the inverse-based fractional-power domain and the one-line antiderivative. If density was silently intended, reclassify the result as a sharp obstruction: it proves that density is indispensable, while leaving the dense-domain question open.

References

- [1] C. Batty and C. Chen, *Besov spaces associated with non-negative operators on Banach spaces*, arXiv:2006.07008v1 (2020).